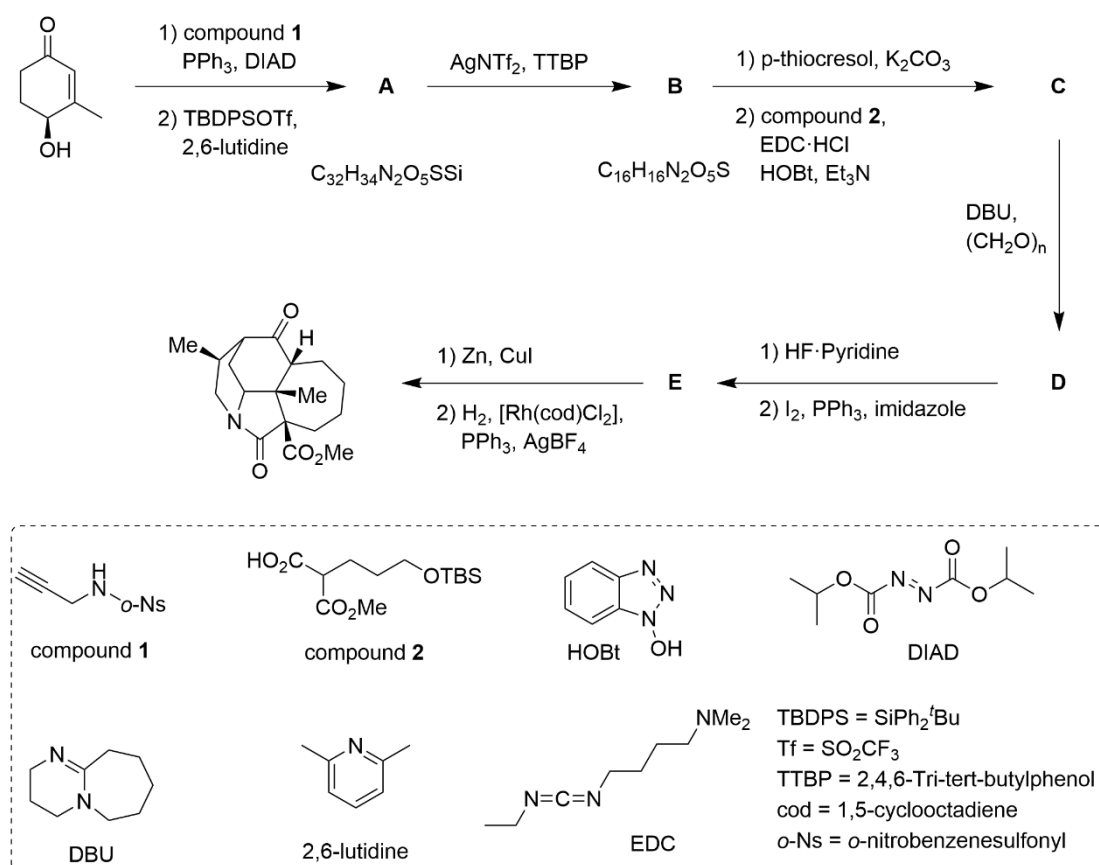


Problem 5 Daphniphyllum alkaloids

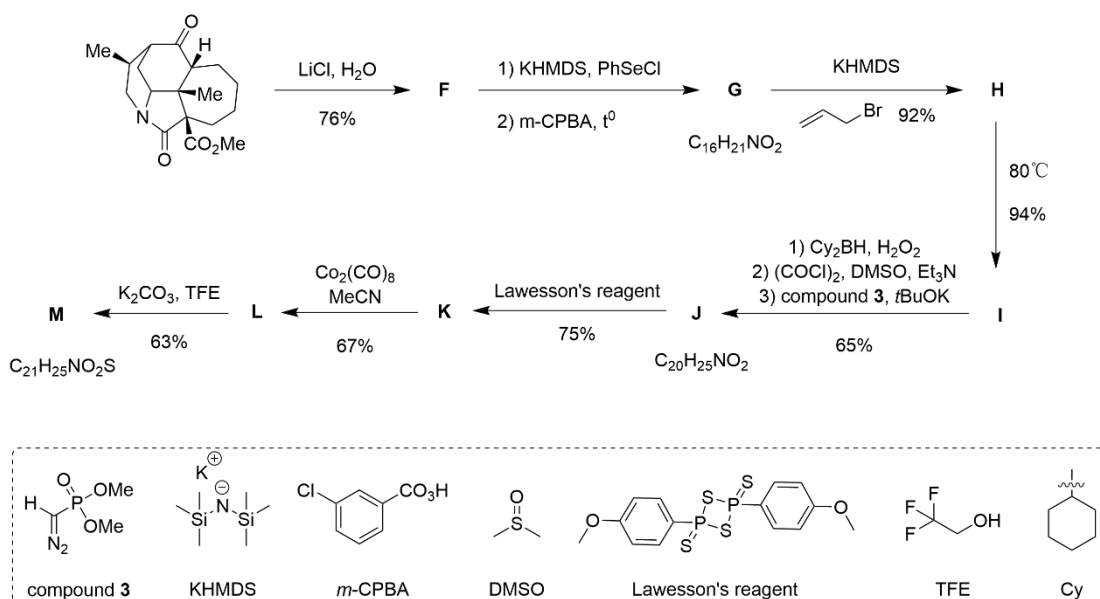
One group of bioactive alkaloids that fight cancer and viruses is the Daphniphyllum alkaloids. The following scheme shows the synthesis of a fragment that serves as their important structural backbone:



5.1 Draw the structures of compounds **A** – **E**, **B'** with indicating stereochemistry.

Hint: In **A** → **B** process bicyclic compounds **B** and **B'** are formed.

One of the key reactions in the synthesis of Hybridaphniphylline B (**V**), a member of this alkaloid group, is the Diels-Alder reaction. The following scheme shows the first steps in the synthesis of the diene (**O**) fragment:



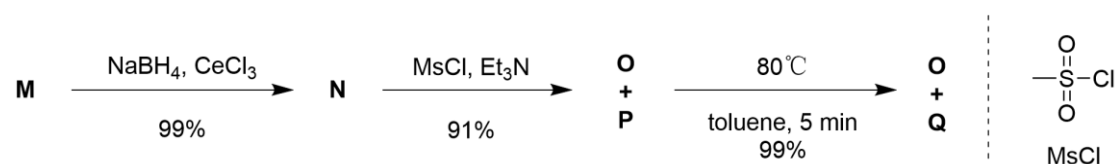
5.2 Draw the structures of compounds **F** – **M**, **I'** with indicating stereochemistry.

Hints:

- * The **H** compound is formed via dienolate.
- * In the **H** $\rightarrow$ **I** process, the **I'** isomer is also formed along with **I**, and compound **I** contains one more quaternary carbon atom than **I'**.
- * The **K** $\rightarrow$ **L** process is Pauson-Khand reaction, which is $[2+2+1]$ cycloaddition.
- * The **L** $\rightarrow$ **M** process is isomerization under basic conditions, and the driving force of the reaction is the formation of a more stable isomer.

5.3 The solid state $\text{Co}_2(\text{CO})_8$ used in the **K** $\rightarrow$ **L** transition shows the following signals in the IR spectrum: 2112, 2071, 2059, 2044, 2031, 2001, 1886, 1857 cm^{-1} , and the complex obeys the 18-electron rule of Sidgwick. Draw the structure of this complex.

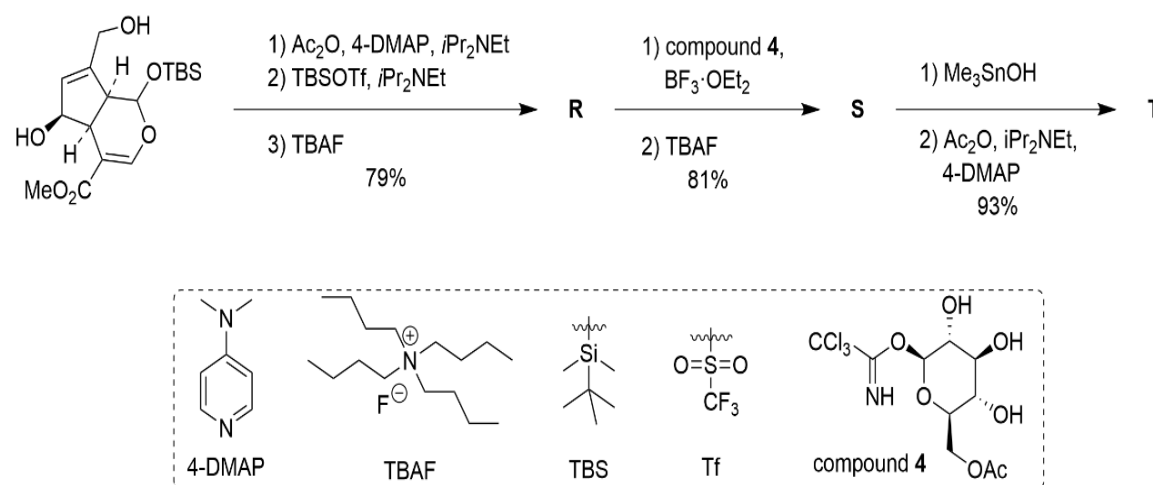
To synthesize the diene **O**, the **M** compound was reacted with NaBH₄ in the presence of CeCl₃. Then compound **N** was reacted with MsCl, Et₃N, a 1:16 mixture of **O** and **P** isomers was formed – it was observed that a certain part of **P** was converted to **O** when the mixture was kept for a long time. If the initial **O** and **P** mixture was heated in a toluene solution at 80 °C, a 2.6:1 mixture of **O** and **Q** isomers was obtained.



5.4 Draw the structures of compounds **N** – **Q**.

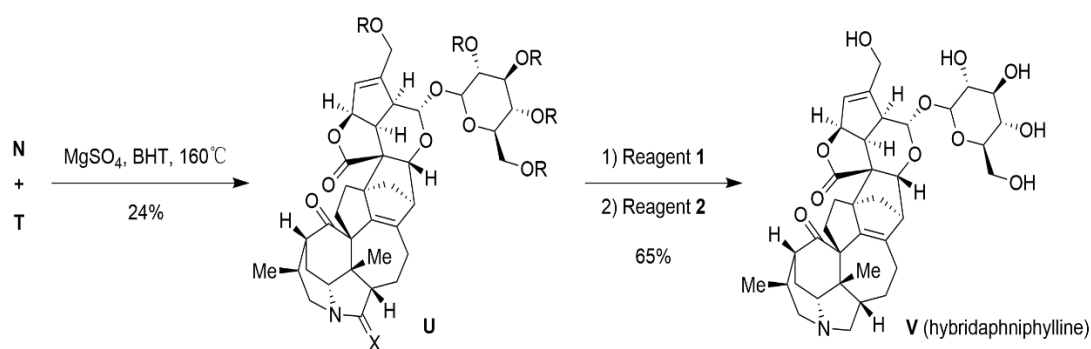
- * In the **N** compound, the number of stereogenic atoms with R- and S- configurations is equal, and the newly formed center has the R- configuration.
- * Among the isomers, only substance **P** contains 7 chiral centers.

Below is another important building block of the synthesis – the formation of the dienophile (**T**) fragment:



5.5 Draw the structures of compounds **R** – **T** with indicating stereochemistry.

Due to the low yield of **U** obtained from the reaction of substances **O** and **T**, substance **O** was formed in-situ from substance **N**:



BHT - 2,6-Di-tert-butyl-4-methylphenol

5.6 Suggest Reagent 1 and Reagent 2 for the **U** $\rightarrow$ **V** transition.

Note: Reagent 1/ Reagent 2 are interchangeable.

5.7 Approximately 100 mg (0.162 mmol) of **V** was formed in this synthesis.

If it is known that **N** and **T** were added in a 1:1 molar ratio, calculate the amounts (mmol) of starting materials used in the syntheses in 5.2 and 5.5.

Note: The yields of the reactions are shown.